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 Studierichting: Informatica
 Oefeningenreeks 2D nr. 6.8

$$f(x) = \frac{5x + 1}{9x^2 - 4}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{5x + 1}{9x^2 - 4} \right) \\ = \frac{5 \cdot 0 + 1}{9 \cdot (0)^2 - 4} = -\frac{1}{4} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow -1} \left(\frac{5x + 1}{9x^2 - 4} \right) \\ = \frac{5 \cdot (-1) + 1}{9 \cdot (-1)^2 - 4} = -\frac{4}{5} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 1} \left(\frac{5x + 1}{9x^2 - 4} \right) \\ = \frac{5 \cdot 1 + 1}{9 \cdot 1^2 - 4} = \frac{6}{5} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow -\frac{2}{3}} \left(\frac{5x + 1}{9x^2 - 4} \right) \\ = \frac{5 \cdot -\frac{2}{3} + 1}{9 \cdot \left(-\frac{2}{3} \right)^2 - 4} \\ = \frac{-\frac{10}{3} + \frac{3}{3}}{\frac{36}{9} - 4} = \frac{-\frac{7}{3}}{0} \end{aligned}$$

$$\lim_{x \rightarrow -\frac{2}{3}-} f(x) = -\infty$$

$$\lim_{x \rightarrow -\frac{2}{3}+} f(x) = +\infty$$

References